

Supplementary Materials

Appendix A. Knowledge-Foraging Task: Instructions

The instructions for the low-creativity condition are as follows: “Your task is to create an explanation that answers the following questions. The following questions can be fully answered by information found on Wikipedia. You will need to find the information to answer the questions by searching for the information using the Wikipedia window located on the right side of the screen. We ask that you record your explanation in the textbox on the left side of the screen.”

The instructions for the high-creativity condition are as follows: “Your task is to create an explanation that answers the following questions. The following questions cannot be fully answered by information found on Wikipedia. Instead, you will need to combine or synthesize information found while searching Wikipedia to construct an answer. You can search for the information using the Wikipedia window located on the right side of the screen. We ask that you record your explanation in the textbox on the left side of the screen.”

Participants also received the following general instructions for both conditions:

- Each explanation should be supported by information from multiple Wikipedia pages.
- Your explanation must be at least 3 sentences long, but we encourage you to provide longer explanations if it is needed to fully answer the question.
- Use complete sentences when answering the question.
- You can switch to the Wikipedia browser to look for information at any time.
- While developing your explanation, you are allowed to copy and paste information found on Wikipedia into the textbox.

Participants were also asked to report their familiarity with the topic (before answering each question) using a 7-point Likert scale, where 1 indicates “Not Familiar” and 7 indicates “Extremely Familiar.” The familiarity ratings are not included in the analysis for this manuscript.

Appendix B. Knowledge-Foraging Task: Stimuli

Practice Stimulus.

1. What do Capybaras eat?

Stimuli from the Low-Creativity Condition.

1. How do species survive exceptionally harsh conditions? (ecology)
2. How are mental/psychological disorders diagnosed? (psychology)
3. How have people made red dye during ancient and modern times? (art history)
4. What are some of the causes that led to economic depressions? (economics)

Stimuli from the High-Creativity Condition.

1. How is evolution impacted by species becoming increasingly connected through globalization? (ecology)
2. What brain mechanisms help people make scientific discoveries? (psychology)
3. How do ancient works of art play a role in contemporary society? (art history)
4. What caused income to vary between ethnic groups? (economics)

Appendix C. Preprocessing Wikipedia Pages

Before running the analysis we preprocessed the Wikipedia pages visited by the participants. First, we excluded search results pages. For the *condition networks* we also removed duplicate pages, where a participant visited the same page multiple times or when participants click on subheader links on a Wikipedia page. Additionally, we dropped 105 pages due to a hyperlink error (where the link went to a different page than the one mentioned), 22 pages because the Wikipedia page was blank, 20 disambiguation pages that displayed links to various pages with the same name, and 2 Wikipedia pages that only contained lists without complete sentences.

Next, we scraped text from the Wikipedia pages. Specifically, we extracted plain text, glossary terms, and bullet points that were complete sentences. We then cleaned the extracted text by removing footnote superscripts, error messages, notices, duplicate paragraphs, and leading or trailing white spaces.

Appendix D. Preprocessing Participants' Responses

Before running our analysis, we preprocessed the participants' responses to the questions. First, we removed instances where the participant took the study multiple times. For these cases, we only kept the participants' responses when they first took the study.

Next, we processed the participants' responses to the questions. In our task, we recorded both the final response to each stimulus and any intermediate responses made when participants edited their answers. Specifically, we logged each response when a

participant moved the cursor outside the textbox, where they had entered their answer. This movement indicated that they may have updated the text. There were 16 instances where the final response was not sent to Matomo, an online platform that we used to record our data (Matomo, n.d.). For these 16 instances, we were able to use the last intermediate responses recorded.

Additional preprocessing steps for specific analyses are provided in Appendix E and Appendix F.

Appendix E. Semantic-Similarity Measures

Term-Frequency Inverse-Document Frequency (tf-idf). For our study, we used term-frequency inverse-document frequency (tf-idf) to identify the semantic similarity between documents (i.e., between the text from pairs of Wikipedia pages and between the participants answers and the text from the Wikipedia pages). Tf-idf identifies words in a document that distinguish it from a set of documents; by pinpointing words that are semantically relevant to a document, while downweighting words that are commonly used throughout a set of documents (Salton & Buckley, 1988). The procedure results in a vector of tf-idf scores for every document, which include a score for each distinct word present within a set of documents. Words that are assigned a high value help distinguish that document from other documents in the set. For our analysis, we compared the semantic similarity between pairs of documents (represented as tf-idf vectors) using cosine similarity, which measures the angle between the vectors. High values of cosine similarity (on a scale from 0 to 1) indicate that the two documents have similar tf-idf vectors, and are semantically similar to each other.

In this study, we used tf-idf in multiple ways. First, we used it to compare the semantic similarity between Wikipedia pages. That is, we found the cosine similarity between tf-idf vectors derived using the text from Wikipedia pages. The resulting semantic similarity was used as the edge weights for our networks.

Next, we used tf-idf to calculate the semantic distance between the participants' answers and the text from the Wikipedia pages. Specifically, we calculated the cosine similarity between the vector for the text from the participants' answer and the vector for the text from each individual Wikipedia page that the participant viewed while typing their answer. Our analysis was conducted on the average semantic similarity between the participants' answer and the Wikipedia text for each condition.

Sentence Transformers. We used Sentence Transformers to measure the semantic similarity between the sentences from the participants' answers and the sentences from the Wikipedia pages. Sentence Transformers provide semantically meaningful sentence embeddings that can be used to calculate the semantic similarity between pairs of sentences (Reimers & Gurevych, 2019). For our analysis, we used the 'sts-b-roberta-large' model, which is a version of the RoBERTa model that is well-suited

for assessing semantic similarity between sentences (Y. Liu et al., 2019). We first created embeddings for each sentence from the participants’ responses and for every sentence from the Wikipedia pages visited by the participants. We then compared the meaning of pairs of sentences by finding the cosine similarity of pairs of sentence embeddings.

For our analysis, we identified the sentence from the Wikipedia pages that was the most similar to or “best match” for each sentence in the participants’ answer. We used the score associated with the best match to calculate the average similarity between the participants sentences and the best matching sentences from the Wikipedia pages for each condition.

Latent Dirichlet Allocation (LDA). We also sought to compare and contrast the topics mentioned in the participants answers with the topics mentioned in the Wikipedia text. To do this, we employed a topic modeling technique called Latent Dirichlet Allocation (LDA), which is a generative probabilistic model used to identify latent or hidden topics from a set of documents (Blei, Ng, & Jordan, 2003). In LDA, a topic is a distribution over words. The words are weighted based on how well they map onto a particular topic. Each document is associated with a list of topics, wherein each topic is assigned a probability of being in a document. The probability of a topic being in a document is based on the distribution of words in the document, and follows the Dirichlet distribution.

We ran our analysis per stimulus, where we identified topics associated with both the participants’ answers and the Wikipedia pages visited by all participants for that stimulus. To see how well the topics from the answers mapped onto the topics from the Wikipedia pages, we found the correlation between the topic distribution for the participants’ answers and the topic distribution for each Wikipedia page visited by the participant.

Appendix F. Preprocessing Latent Dirichlet Allocation (LDA)

For our LDA analyses, we ran separate analyses for each stimulus. That is, the corpus only included the participants responses to a particular stimulus and the Wikipedia pages participants viewed in response to that stimulus. We began by preprocessing the text by lower casing the letters, removing punctuation and stop words, tokenizing the words, and lemmatizing each token. Next, we selected the number of topics for each stimulus based on the coherence score, which indicates whether the words in a topic are more semantically similar (Röder, Both, & Hinneburg, 2015). We calculated the coherence score for a range of topics (between 2 and 50), and plotted the curve of these results. Next, we used the knee approach to identify the best number of topics. This approach identifies the point in the plotted curve where increases in the coherence score are minimal and insignificant (Satopaa, Albrecht, Irwin, & Raghavan, 2011). We then ran our topic models using the identified optimal number of topics.

Appendix G. Pilot Study

For this study, we recruited 64 undergraduates from a large Midwestern university (41 females, Mean age = 30.09). The version of the Knowledge-Foraging Task differed from the one presented in the main manuscript in that the participants had only five minutes to answer the questions. They were also presented with 2 additional stimuli per condition, for a total of eight stimuli.

For the analysis, we found that the high-creativity condition had larger average clustering coefficients than the low-creativity condition ($p < 0.001$, Effect size = -0.47). We also found that the average semantic similarity between the participants' answers and Wikipedia pages was lower for the high-creativity condition than the low-creativity condition ($p < 0.001$, Effect size = -0.59).

Appendix F. GPT Classification Procedure

To identify responses that did not attempt to address the question, we used the OpenAI API with the GPT-5 model (accessed September 10, 2025). The model was queried using the following system prompt:

“Your task is to classify answers as either 1 or 0. Respond 1 if the text does not attempt to answer the question or is completely off-topic. Respond 0 if the text attempts to answer the question, even if the answer is untrue. If the text starts with something like ‘I don’t know’ but then provides relevant information that addresses the question, respond 0. Output only 1 or 0 with no explanation.”

For each trial, the model received the question and the participant’s answer in the following format: “Question: ‘question’ Answer: ‘answer’”

All calls used the default OpenAI API settings, including a temperature of 1.0 (no temperature parameter was modified in the API request). No other sampling or decoding parameters were manually altered.